Deep Learning in Stroke Care

1 Introduction

Stroke is known as a leading cause of disability and mortality worldwide, affecting millions of individuals each year with cases increasing due to aging populations and risk factors like high blood pressure and diabetes Feigin et al. [2017]. It occurs when the blood supply to the brain is interrupted due to either an ischemic blockage or a hemorrhagic event, leading to the rapid loss of neurological function. The timely and accurate diagnosis of stroke is essential, as early intervention significantly improves patient outcomes and reduces long-term complications. However, conventional diagnostic methods, such as Computed Tomography (CT) scans and Magnetic Resonance Imaging (MRI), require expert radiological interpretation, which can be time-consuming and subject to variability among clinicians. Furthermore, the availability of stroke specialists is often limited, especially in resource-constrained settings, leading to delays in diagnosis and treatment. These challenges necessitate the development of more efficient, accurate, and accessible diagnostic solutions to improve stroke care globally. Healthcare systems face the significant challenge of providing quality care while controlling rising costs. In this situation, advanced technologies, especially deep learning as a part of artificial intelligence, offer the potential to improve stroke care, from diagnosis to rehabilitation Litjens et al. [2017].

Artificial Intelligence (AI) has emerged as a transformative tool in medical imaging and diagnosis, offering the potential to address the challenges faced in stroke detection and assessment. Machine learning and deep learning models can analyze complex data with a high degree of accuracy, facilitating rapid and automated stroke detection. The potential of deep learning in healthcare is substantial, as it leverages large volumes of data to uncover patterns and insights that might be difficult to detect otherwise Esteva et al. [2017]. In stroke care, it can transform various stages of management—from rapid and precise diagnosis to the development of personalized treatment plans and the prediction of recovery or recurrence Perry et al. [2020]. AI-driven systems can assist radiologists by identifying subtle signs of stroke Cai et al. [2022], differentiating between ischemic and hemorrhagic strokes Gautam and Raman [2021], and even predicting patient outcomes based on imaging and clinical data Liu et al. [2023]. Beyond imaging, AI algorithms are being integrated into Electronic Health Records (EHRs) and wearable technologies to detect early warning signs of stroke, thereby enabling preventive interventions Olawade et al. [2024]. By employing AI in stroke diagnosis, healthcare providers can enhance diagnostic accuracy, reduce interpretation time, and extend expert-level analysis to underserved regions. However, integrating these advanced technologies into clinical practice is not without its own set of challenges. Critical issues such as ensuring data quality, protecting patient privacy, and navigating the ethical implications of using AI in medicine must be addressed with careful consideration and robust frameworks Char et al. [2018]. This chapter explores the various applications of deep learning in stroke care, offering an overview of current uses and future possibilities. It also provides a foundation for understanding how these technologies can help diagnose, treat, and predict outcomes in stroke patients by highlighting both their potential and the hurdles that must be overcome.

1.1 Stroke: Definition, Types, and Progression

Stroke is medically defined as the sudden disruption of blood supply to the brain, resulting in rapid neuronal damage due to oxygen and nutrient deprivation. It is the second leading cause of mortality worldwide and the third leading cause of long-term disability, with consequences such as paralysis, speech difficulties, and cognitive impairments affecting millions of individuals Feigin et al. [2022]. Beyond its direct health implications, stroke profoundly affects families and communities by increasing caregiving demands and reducing productivity. The economic burden is substantial, with healthcare systems facing significant costs related to treatment, rehabilitation, and long-term care. The importance of early detection and individualized treatment planning cannot be overstated, as timely intervention is critical to minimizing brain damage and improving long-term outcomes. Given these far-reaching consequences, early diagnosis, effective treatment, and rehabilitation strategies are essential to mitigating stroke-related morbidity and mortality. Clinically, stroke is categorized into three primary types:

- **Ischemic Stroke:** Accounting for the majority of stroke cases, ischemic stroke occurs when a blood vessel supplying the brain becomes obstructed by a clot or embolus. This blockage can originate locally in the brain's arteries (thrombotic stroke) or travel from another part of the body (embolic stroke).
- **Hemorrhagic Stroke:** Less common but often more severe, hemorrhagic stroke results from the rupture of a blood vessel in the brain, leading to intracranial bleeding, increased pressure, and subsequent neuronal damage. This condition is further classified into: Intracerebral Hemorrhage (ICH), in which bleeding occurs within brain tissue, and Subarachnoid Hemorrhage (SAH), in which bleeding occurs in the space surrounding the brain, often due to a ruptured aneurysm.
- **Transient Ischemic Attack (TIA):** Often referred to as a "mini-stroke," TIA involves a temporary reduction in blood flow to the brain, with symptoms resolving within 24 hours. While it does not cause permanent damage, TIA serves as a critical warning sign of a potential future stroke, necessitating prompt medical intervention.

Figure 1 illustrates the fundamental difference between ischemic and hemorrhagic strokes. With an understanding of stroke types and their clinical progression, it becomes essential to explore how emerging technologies can enhance current practices. Stroke progression is classified into four stages, each characterized by distinct pathophysiological changes and clinical objectives: hyperacute, acute, subacute, and chronic.

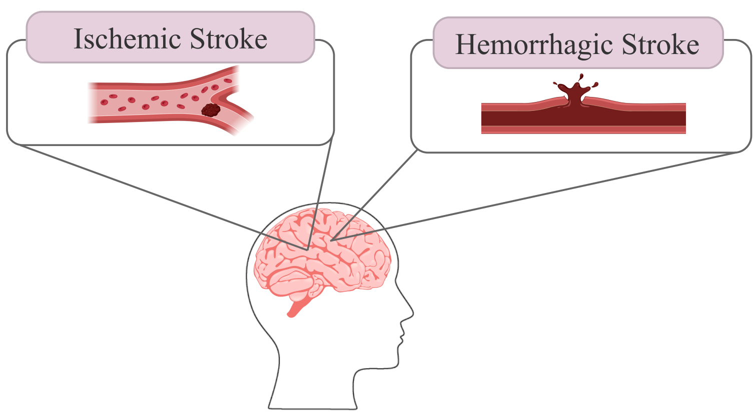

Figure 1: Different stroke types that can cause permanent damage. Ischemic stroke occurs due to an obstruction in a blood vessel and Hemorrhagic one occurs due to the rupture of a blood vessel.

- **Hyperacute Stage (Within the First 8 Hours)**: This phase is marked by an abrupt cessation of cerebral blood flow, triggering rapid neuronal injury. Immediate intervention is essential to preserve viable brain tissue. Treatment strategies focus on thrombolysis (for ischemic stroke) or hemorrhage control (for hemorrhagic stroke).
- **Acute Stage (Hours to a Few Days)**: During this stage, ischemic changes become evident on imaging, and clinical symptoms such as hemiplegia and language deficits manifest. The primary goal is to prevent stroke progression and assess the efficacy of initial treatment.
- **Subacute Stage (Days to Weeks, Typically 2–6 Weeks)**: The brain begins its healing process, and symptoms stabilize. Early rehabilitation plays a critical role in functional recovery, emphasizing motor and cognitive therapy.

- **Chronic Stage (Beyond Several Weeks)**: At this stage, the infarcted brain tissue is fully developed, and neurological deficits become more stable. Long-term rehabilitation and stroke prevention strategies are prioritized to enhance the patient's quality of life and reduce the risk of recurrence.

A comprehensive understanding of stroke types and progression is essential for developing effective diagnostic, therapeutic, and rehabilitation approaches. With advancements in AI, particularly deep learning, these insights can be leveraged to enhance stroke diagnosis, prognosis prediction, and personalized treatment strategies, ultimately improving patient outcomes.

1.2 Current Clinical Practices

Despite significant advancements in stroke care, several limitations persist in current clinical practices that impact both patient outcomes and healthcare efficiency. One of the primary challenges is the timeliness of diagnosis. Time is critical in stroke treatment, particularly for ischemic strokes where, as emphasized by the clinical principle 'time is brain.' In current clinical practice, when a patient presents symptoms such as sudden weakness, difficulty speaking or facial drooping, emergency medical teams follow the FAST (Face, Arms, Speech, Time) protocol for quick assessment. Once a stroke is suspected, the patient undergoes neuroimaging to differentiate between ischemic and hemorrhagic strokes. In addition, blood tests, Electrocardiograms (ECG) and carotid ultrasounds are used to identify risk factors such as clotting disorders, atrial fibrillation, or carotid artery stenosis. This traditional approach often requires significant time for setup, imaging, and interpretation, which can delay treatment initiation.

Accuracy in diagnosis remains also a challenge, with imaging modalities sometimes failing to pinpoint the exact nature or location of a stroke, especially in the early stages. This can lead to misdiagnosis or delayed treatment, both of which are detrimental to patient outcomes. In addition, variability in human interpretation of imaging results can introduce subjectivity, reducing the consistency and reliability of diagnoses in different healthcare settings. Moreover, there is the issue of resource allocation. Stroke care, especially in the acute phase, is resource-intensive, requiring specialized personnel, equipment, and facilities which are not universally available, particularly in rural or economically disadvantaged area. This disparity can lead to unequal access to quality stroke care, influencing recovery disparities.

Rehabilitation and long-term care post-stroke also face limitations. In the acute phase, early mobilization is encouraged to prevent complications like pneumonia and deep vein thrombosis. In the subacute stage, intensive rehabilitation begins with a multidisciplinary team that includes physical therapists, occupational therapists, and speech-language pathologists to help restore motor skills, daily activities, and communication abilities. Long-term rehabilitation in the chronic phase continues based on the patient's needs. Current practices often rely on standardized protocols that do not fully account for the unique needs of each patient, leading to less effective recovery processes. Lack of personalized rehabilitation plans can prolong recovery time and affect quality of life after a stroke.

Stroke survivors are at high risk of recurrent strokes, so predicting and preventing another event is a key clinical goal. Risk assessment models, such as the ABCD2 score (which considers Age, Blood pressure, Clinical symptoms, Duration, and Diabetes status), help identify high-risk patients. While clinical scores and scales exist, they do not always incorporate the nuanced data that could predict individual patient trajectories more accurately, thus missing opportunities for preventive care or targeted interventions.

1.3 The Role of Deep Learning for Stroke Care

Deep learning, a subset of artificial intelligence, has emerged as a powerful and transformative technology in various domains, including healthcare. Using vast amounts of data and advanced architectures, deep learning models learn in a data-driven manner, enabling them to identify complex patterns and make informed decisions. In medical applications, this capability facilitates fast and accurate analysis, improving diagnostic precision and clinical workflows. For stroke, where timely intervention is critical, deep learning holds significant potential in all stages of care, from early detection and diagnosis to treatment planning and prognosis prediction. By improving efficiency and accuracy, deep learning-based approaches can contribute significantly to better patient outcomes. One of the key strengths of deep learning is its ability to process and analyze diverse types of data, including neuroimaging modalities (e.g., CT, MRI), electronic health records, and real-time physiological signals. For instance, Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in automating the detection of ischemic and hemorrhagic strokes in medical images, as well as segmenting affected areas from healthy tissue, significantly reducing the time required for diagnosis. Beyond imaging, Recurrent Neural Networks (RNNs) and transformer-based models have been employed for predicting patient outcomes and optimizing treatment strategies. Deep learning also plays an essential role in stroke rehabilitation, where models can analyze movement patterns from video or sensor data and model temporal sequences in patient recovery trajectories, helping tailor personalized rehabilitation plans.

However, integrating deep learning models into clinical workflows comes with several challenges that must be addressed to ensure trustworthiness and seamless adoption. A trustworthy AI system must exhibit transparency, meaning its decision-making processes should be explainable, and any errors must be traceable with clear lines of accountability. Fairness is another critical factor, as models must be developed to be unbiased and inclusive, accounting for diverse patient populations and guarding against the amplification of existing health disparities. Reliability is equally important; AI models must perform consistently well across various clinical scenarios and remain robust against anomalies, such as imaging artifacts or unexpected data variations. Additionally, given that AI tools often require access to sensitive health information, data security and patient privacy are paramount, necessitating models designed with privacy-preserving mechanisms to mitigate risks. Finally, for these systems to be successfully adopted, effective human-AI interaction is essential. This includes providing clear confidence scores for AI predictions and offering intuitive visualization tools that allow for quick verification of the model's findings. Such interfaces must integrate smoothly into existing clinical workflows, enabling clinicians to easily review, interpret, and ultimately trust the AI-generated results. Addressing these multifaceted challenges is fundamental for the successful and ethical deployment of AI-driven systems in stroke diagnosis and treatment.

2 Imaging Modalities in Stroke Diagnosis

Medical imaging is an essential tool for timely and accurate stroke diagnosis, enabling clinicians to select appropriate treatment strategies. The primary imaging modalities used for stroke evaluation include CT and MRI, along with specialized techniques such as CT Angiography (CTA), CT Perfusion (CTP), Diffusion-Weighted Imaging (DWI), Apparent Diffusion Coefficient (ADC), Perfusion-Weighted Imaging (PWI), and Magnetic Resonance Angiography (MRA). Each modality offers unique advantages and limitations, influencing its role in clinical decision-making. The choice between CT and MRI, as well as their specialized applications, depends on factors such as clinical urgency, imaging availability, and patient-specific considerations.

2.1 Computed Tomography (CT)

CT is often the first-line imaging modality for stroke assessment due to its speed, availability, and cost-effectiveness. It is particularly effective in rapidly identifying hemorrhagic stroke and large ischemic infarcts, which is essential for determining immediate treatment options. However, Non-Contrast CT (NCCT) has limited sensitivity for detecting early ischemic changes or smaller infarcts, especially in the first few hours after stroke onset Lin and Liebeskind [2016]. An example of a CT scan showing a stroke lesion is provided in Figure 2. Additionally, CT imaging involves exposure to ionizing radiation, which remains a drawback. Advanced CT techniques such as CT Angiography and CT Perfusion can enhance the diagnostic capabilities. CT Angiography utilizes contrast agents to visualize vascular occlusions, aiding in patient selection for thrombolytic therapy or mechanical thrombectomy Lin and Liebeskind [2016]. CT Perfusion assesses cerebral blood flow dynamics, allowing differentiation between recoverable brain tissue (penumbra) and irreversibly damaged tissue (core infarct). CTPs have some drawbacks, including the complexity of interpretation, which often requires validated software. However, different software solutions may generate varying outputs using different approaches, leading to inconsistencies. In addition, CTP struggles to detect reperfused infarcted tissues, and its output can be affected by patient motion Christensen and Lansberg [2019].

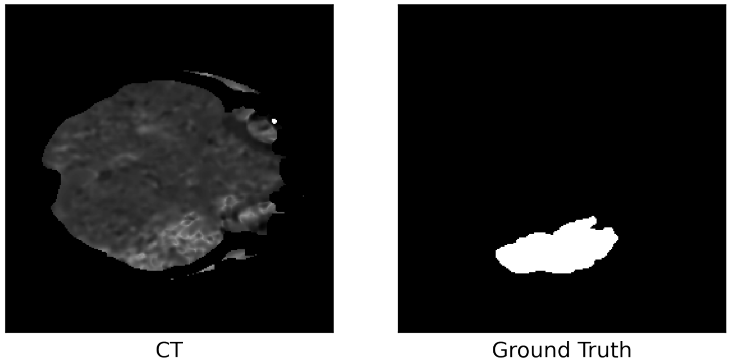

Figure 2: Sample of stroke infarct on CT images with the annotation, from Cereda et al. [2016], Hakim et al. [2021].

2.2 Magnetic Resonance Imaging (MRI)

MRI provides higher soft tissue contrast and is particularly effective for detecting ischemic changes earlier than CT, without exposing the patient to ionizing radiation. However, MRI is less widely available, time-consuming, and contraindicated in some patients with implanted medical devices or severe claustrophobia. MRI employs strong magnetic fields and radiofrequency pulses to generate detailed anatomical and functional images of the brain. Different MRI sequences provide distinct insights into stroke pathology.

DWI is highly sensitive to acute ischemia, detecting stroke-related changes within minutes of onset. It highlights restricted water movement caused by cytotoxic edema, providing a clearer picture of stroke extent and location Lin and Liebeskind [2016]. ADC imaging enhances DWI by providing a quantitative measure of water diffusion in brain tissue, helping to distinguish truly infarcted regions and it provides parameters for outcome prediction in acute ischemic patients Van Everdingen et al. [1998]. One key advantage of ADC imaging is its ability to eliminate T2 shine-through effects—situations where high signal intensity in DWI is mistakenly attributed to restricted diffusion when, in reality, it results from an underlying high T2 signal. Figure 3 provides a visual comparison of a stroke infarct on DWI and ADC images. PWI evaluates cerebral hemodynamics, including Cerebral Blood Flow (CBF) and Cerebral Blood Volume (CBV) Simonsen et al. [2015]. PWI is instrumental in identifying penumbral tissue, which may be recoverable with timely intervention.

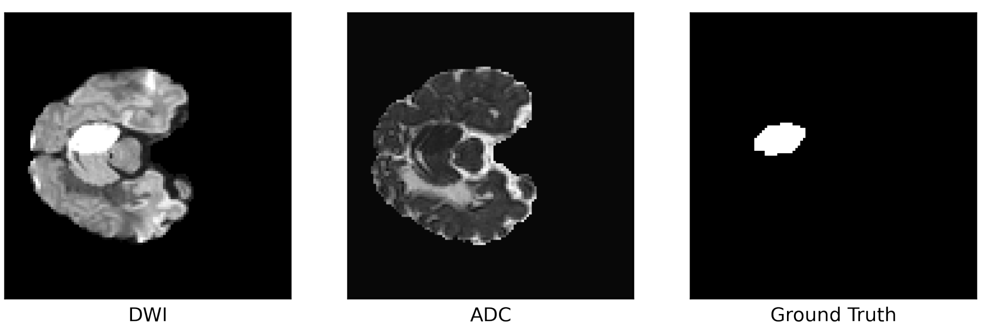

Figure 3: Sample of stroke infarct on Diffusion-weighted and Apparent diffusion coefficient MRI with the annotation, from Hernandez Petzsche et al. [2022]. The infarct appears hyperintense in DWI and hypointense in ADC.

T1-Weighted (T1W) Imaging provides detailed anatomical information based on proton relaxation times. It is useful for detecting chronic infarcts and hemorrhagic transformation, but its sensitivity for acute ischemic infarcts is lower compared to DWI and ADC Van Everdingen et al. [1998]. T2-Weighted (T2W) Imaging highlights fluid-containing structures, making it effective for detecting edema and chronic tissue damage. Areas with high water content appear hyperintense (bright) on T2W images. T2W imaging has also lower sensitivity for detecting the acute ischemic lesions compared to DWI Lansberg et al. [2000]. Fluid-Attenuated Inversion Recovery (FLAIR) suppresses signals from Cerebrospinal Fluid (CSF) to enhance visualization of white matter abnormalities. It is particularly useful for detecting small cortical infarcts that may be overlooked on standard T2W images. However, the ability of FLAIR to detect stroke lesions varies over time. In the hyperacute phase of ischemic stroke (within a few hours), FLAIR may not show significant changes, potentially leading to an underestimation of the lesion Kim et al. [2014].

Additionaly, Magnetic Resonance Angiography is a non-invasive imaging modality that provides detailed visualization of cerebral arteries, allowing for the assessment of vascular abnormalities such as stenosis, occlusions, and aneurysms Kim et al. [2014]. Compared to conventional catheter-based angiography, MRA offers reduced procedural risks and greater patient comfort. It employs two primary techniques: 1) Time-of-Flight (TOF) MRA, which captures blood flow by inducing blood inflow effects without contrast agents, 2) contrast-enhanced MRA, which uses contrast material to improve visualization while carrying a lower risk of allergic reactions compared to contrast agents used in CT angiography Graves [1997]. In the TOF MRA, the vascular signal depends on blood flow direction and velocity, making it more susceptible to artifacts and potentially leading to an overestimation of stenosis Czap and Sheth [2021].

Table 1: Summary of different imaging modalities for stroke.

Modality	Best For	Pros	Cons
Non-Contrast CT	First-line emergency imaging, hemorrhage detection	Fast ($\tilde{5}$ min), widely available, detects hemorrhages	Limited sensitivity for acute ischemia, radiation exposure
CT Angiography	Detecting large vessel occlusions	High-resolution vascular imaging, fast, non-invasive	Requires contrast (risk of allergy/kidney damage), radiation exposure
CT Perfusion	Differentiating infarct core vs penumbra	Fast, painless, guides reperfusion therapy decisions	Requires Contrast, inability for detecting the reperfused infarcted tissue, complex interpretation, susceptibility to motion
Diffusion-Weighted Imaging	Most sensitive for acute ischemic stroke	Highest sensitivity for setecting stroke within minutes	May be affected by T2 shine-through
Apparent Diffusion Coefficient	Confirming true infarct	Helps confirming ischemia	Limited ability in accurate detection in very early stroke
Perfusion-Weighted Imaging	Differentiating infarct core vs penumbra	Helps in reperfusion therapy, Good spatial resolution	Requires contrast, complex analysis
T1-Weighted Imaging	Detecting chronic infarcts and hemorrhagic transformation	Good for structural details	Low sensitivity for acute stroke
T2-Weighted Imaging	Detecting edema and chronic infarcts	Detects white matter damage	Low sensitivity for acute stroke
Fluid-Attenuated Inversion Recovery	Differentiating acute vs. chronic infarcts	Suppresses CSF signal, improved small lesion detection	Sensitive to timing
Magnetic Resonance Angiography	Detecting artery occlusion or stenosis	Non-invasive vascular imaging	Overestimates stenosis

Various medical imaging techniques are used in stroke assessment to provide detailed and visual information about a patient's condition. Among them, CT and MRI are the most widely utilized. CT is the first-line imaging modality in emergency settings due to its accessibility and speed, making it particularly valuable for detecting hemorrhagic stroke. In contrast, DWI and ADC mapping in MRI are considered the gold standard for early ischemic stroke detection, offering superior sensitivity compared to other techniques. Additionally, CTP and PWI are instrumental for differentiating the infarct core from the penumbra. CT angiography and MR angiography provide essential visual information for detecting arterial occlusions. Understanding these imaging modalities and their applications is fundamental for developing deep learning-based pipelines in stroke care. A summary comparison of these imaging techniques is presented in Table 1.

3 Deep Learning in Stroke

Deep learning has revolutionized stroke care by offering advanced capabilities in diagnosis, prognosis, and rehabilitation. CNNs and transformer-based models have demonstrated remarkable proficiency in analyzing medical images, enabling automated detection of stroke-related abnormalities with high accuracy Zafari-Ghadim et al. [2024]. These models can distinguish between ischemic and hemorrhagic strokes, assess infarct volume, and detect early signs of stroke that might be missed by human observers. Beyond diagnosis, deep learning plays a significant role in prognosis by predicting patient outcomes based on multimodal data, including imaging, clinical history, and biomarkers. Such predictive models help clinicians make informed decisions about treatment strategies, such as thrombectomy or thrombolysis, and estimate the likelihood of recovery. Furthermore, deep learning is increasingly being applied in rehabilitation, where AI-powered tools personalize therapy by analyzing patient progress, optimizing rehabilitation exercises, and even utilizing Brain-Computer Interfaces (BCIs) to assist motor function recovery. More details on the role of deep learning are discussed in this section.

3.1 Diagnosis

Deep learning is a powerful tool for stroke diagnosis, enabling rapid and automated analysis of CT and MRI scans to detect both ischemic and hemorrhagic lesions with high accuracy. Various architectural approaches, such as CNNs, are employed to segment and classify stroke-related abnormalities, significantly reducing diagnosis time and supporting timely clinical intervention. Beyond detection, these models can quantitatively assess lesion volumes and other critical properties, such as estimating the age of a stroke, thereby enhancing prognostic evaluations and contributing to personalized treatment strategies Mainali et al. [2021].

3.1.1 Task-Specific Deep Learning Approaches

In stroke diagnosis, deep learning models can be designed for various tasks. The most common approaches include classification, lesion detection, and lesion segmentation from medical images. Each task presents unique challenges and considerations that must be carefully addressed when developing deep models to ensure accurate and reliable performance.

Stroke classification using deep learning involves assigning labels to 3D medical images (e.g., CT or MRI scans) or individual 2D slices within a 3D volume. In a 3D classification task, an entire brain scan may be categorized as "ischemic stroke," "hemorrhagic stroke," or "normal," with models like 3D ResNet or 3D DenseNet effectively capturing spatial relationships across the entire volume. Unlike their 2D counterparts that process images slice by slice, 3D models analyze the entire volumetric scan at once, allowing them to better capture spatial relationships across adjacent slices. Alternatively, a per-slice classification approach labels each 2D slice based on the presence or absence of stroke-related abnormalities. In this case, 2D models such as ResNet or EfficientNet process slices individually, with final predictions determined either independently or through aggregation techniques like majority voting, where the final class is determined by the most frequent prediction across all slices.

Cross-entropy loss is commonly used for classification tasks, but given the class imbalance in stroke datasets—where stroke-positive slices are often underrepresented—techniques such as weighted cross-entropy or focal loss can help improve model performance. In summary, cross-entropy loss measures the divergence between predicted and actual class distributions, and modifications like weighting or focal loss help address class imbalance by giving more importance to minority classes.

Detection tasks in stroke diagnosis aim to identify and localize stroke-related abnormalities, such as ischemic regions or hemorrhages, within 3D medical images. This can be performed at the slice level (2D) or across the entire volume (3D). Models such as Fast R-CNN, YOLO, or custom deep learning architectures can be employed to detect these abnormalities. The loss function for detection typically combines a term for classification (e.g., cross-entropy loss) to identify abnormalities with a regression loss such as Smooth L1 Loss or IoU Loss to predict bounding box coordinates. Given the critical need for precise localization of small lesions in stroke diagnosis, IoU-based losses, which directly optimize the overlap between predicted and ground-truth bounding boxes, can be incorporated to enhance detection accuracy.

Segmentation tasks in stroke diagnosis focus on delineating stroke-affected regions, such as ischemic penumbra or hemorrhagic areas, from normal tissue at the pixel level in 2D slices or the voxel level in 3D volumes. Models like U-Net and its advanced variations, such as Attention U-Net Oktay et al. [2018] and nnU-Net Isensee et al. [2021], are widely used to achieve a high-precision segmentation. Dice Loss is widely used for stroke segmentation because it directly optimizes the overlap between predicted and ground truth masks, making it particularly effective for imbalanced datasets where stroke regions are often small. A combination of loss functions, such as Dice Loss and Cross-Entropy Loss, is also common. Additionally, advanced loss functions like Tversky Loss or Focal Tversky Loss are particularly useful for emphasizing small stroke lesions, which are critical for accurate diagnosis and treatment planning.

One of the key challenges in stroke segmentation is its multi-instance nature, where multiple lesions of varying sizes may be present within a single 3D volume or 2D slice. Many commonly used loss functions tend to prioritize the segmentation of larger lesions, potentially leading to reduced accuracy in detecting smaller ones Mabrok et al. [2024]. Addressing this bias is essential for improving model performance and ensuring comprehensive stroke assessment.

Some deep learning models are designed to handle multiple tasks, either through sequential processing or a joint learning approach. In the sequential method, separate models are developed for different tasks, where the output of one model influences the input of the next. For example, a classification-segmentation pipeline may first classify an image to determine the presence of an abnormality before passing it to a segmentation model for further analysis Kumar et al. [2020]. However, this approach has drawbacks, as errors from earlier stages can significantly impact subsequent predictions. Alternatively, multi-task learning can be implemented in a joint model, where different tasks are performed simultaneously within the same network Marcus et al. [2023]. In this approach, while outputs are computed separately, shared feature representations improve learning efficiency and overall model performance. By leveraging shared decision-making processes, joint multi-task models enhance feature extraction and reduce the risk of error propagation, making them more robust for complex medical imaging tasks. However, for this approach to be effective, the tasks should have shared underlying structures to ensure meaningful feature learning and robust performance.

3.1.2 Deep Learning Architectures

Convolutional neural networks have been widely used in deep learning models for various medical imaging tasks, including classification, detection, and segmentation. In CNN-based architectures, multiple convolutional layers automatically extract hierarchical features from input data, making them particularly effective for stroke-related applications such as classifying stroke types, detecting lesions, and segmenting affected regions. CNNs excel at capturing local patterns and textures, offering computational efficiency in processing both 2D and 3D images while maintaining robustness to spatial transformations like translation and rotation. However, CNNs have a fundamental limitationthey struggle to capture long-range dependencies due to their inherently local receptive fields. This drawback can hinder their ability to fully understand complex stroke cases where global contextual information is vital. For instance, a small lesion in one part of the brain may be related to vascular abnormalities far from the lesion site, a connection that locally-focused CNNs may not detect. Furthermore, their fixed geometric structures make them less adept at handling variations in the scale and orientation of features without extensive data augmentation.

Several research efforts have been dedicated to enhancing the receptive field of convolutional layers while ensuring that they maintain the ability to capture both local and global information effectively. Various techniques have been proposed to achieve this goal, each leveraging different architectural modifications. One notable approach, presented in Abulnaga and Rubin [2019], incorporates a pyramid pooling module Zhao et al. [2017], which utilizes multiple kernel sizes to aggregate contextual information from different scales. By processing features at varying spatial extents, this method enables the model to extract global information without compromising fine-grained local details. Similarly, Bal et al. [2023] explored strategies that rely on employing different kernel sizes to generate feature maps across multiple receptive fields. This technique allows the model to learn representations from diverse spatial scales, improving its ability to recognize patterns at various levels of abstraction. In another advancement, Liu et al. [2022] introduced an innovative module known as DSE-ResNet, which was strategically placed in the bottleneck layer to enhance inter-channel dependencies. This module plays an essential role in refining feature representations by dynamically recalibrating channel-wise responses. Additionally, this architecture integrated the Poolformer structure Yu et al. [2022] with convolutional neural networks to further enhance the model's capacity to simultaneously capture both localized fine details and broader contextual information.

Attention mechanisms play an essential role in various deep learning models due to their ability to enhance performance by selectively emphasizing the most relevant features while diminishing the influence of less important ones. In general, attention mechanisms can be classified into two primary categories: additive and multiplicative attention. Additive attention determines feature relevance by computing weighted scores using learned parameters. These scores are then applied adaptively to enhance important features. On the other hand, multiplicative attention, encompassing dot-product attention and gating-based strategies such as those employed in Attention U-Net Oktay et al. [2018], modulates features based on their interrelationships. This form of attention is particularly advantageous due to its computational efficiency, as it relies on matrix multiplications rather than explicit parameter learning. Further, attention mechanisms can be structured based on their focus: channel attention methods, such as Squeeze-and-Excitation networks Hu et al. [2018], emphasize the most informative feature channels, allowing the model to refine its representations and suppress irrelevant activations Abramova et al. [2021]. Spatial attention, in contrast, highlights significant regions within an image or feature map, directing computational resources to the most salient areas. Additionally, cross-attention facilitates interactions between multiple input sources, enabling better integration of complementary information from different modalities or perspectives Soliman et al. [2024].

Self-attention Vaswani et al. [2017], the core mechanism of transformers, enables deep learning models to capture long-range dependencies by dynamically computing relationships between different parts of the input. Unlike traditional convolutional layers, which have a fixed receptive field, self-attention allows models to weigh the importance of each element relative to all others, making it highly effective for complex data structures. transformer-based architectures, such as vision transformers (ViTs) Dosovitskiy et al. [2020] and Swin transformers Liu et al. [2021], leverage self-attention to extract global contextual information, improving diagnostic accuracy across different imaging modalities such as CT and MRI.

In general vision transformer architectures, such as ViT, image processing begins with patching, where an input image is divided into fixed-size patches. Each patch is then flattened into a vector and passed through a linear projection layer to map it into a high-dimensional feature space. These patch embeddings are combined with learned positional encodings to retain spatial information, as self-attention alone lacks inherent spatial awareness. The resulting sequence of embeddings is then fed into a standard transformer encoder, which consists of multiple layers of multi-head self-attention and feed-forward networks. This enables the model to capture long-range dependencies and contextual relationships between different patches, making it highly effective for medical image analysis. The general architecture of a Vision transformer is illustrated in Figure 4.

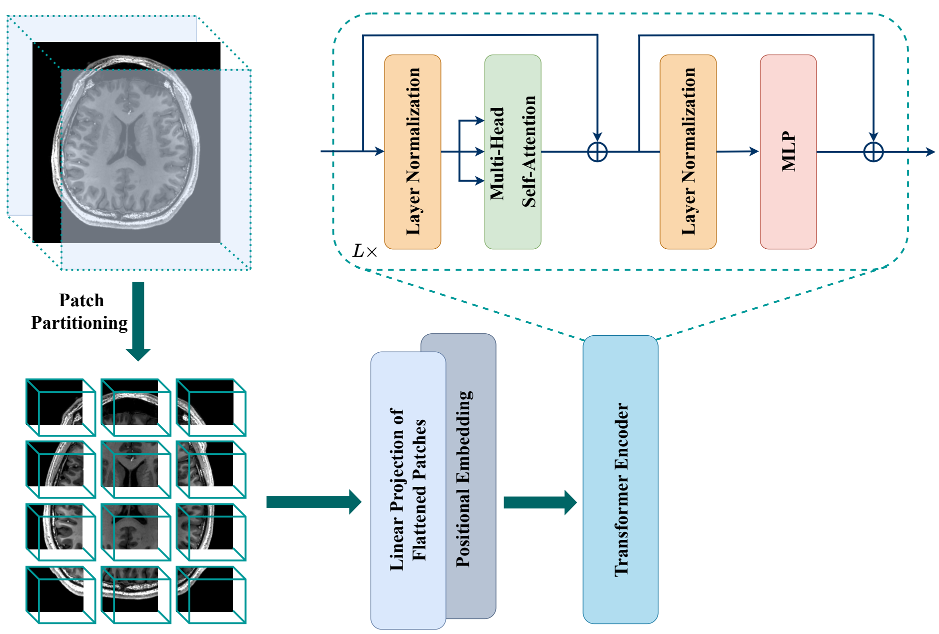

Figure 4: Overview of Vision Transformer (ViT) and the Transformer encoder.

Despite their strong ability to capture long-range dependencies, pure transformers suffer from several limitations, particularly in medical imaging. One major drawback is their lack of inductive biases, such as locality and translation invariance, which are inherent to CNNs. As a result, transformers require extensive amounts of labeled data to generalize effectively. They also struggle with imbalanced feature distributions, a common issue in real-world medical data Soliman et al. [2024]. Additionally, transformers have high computational complexity, as the self-attention mechanism scales quadratically with the number of image patches, making them inefficient for processing high-resolution medical images Dosovitskiy et al. [2020]. To address these challenges, hybrid CNN-transformer architectures have been proposed Alshehri and Muhammad [2024], Ahmed et al. [2024], Wu et al. [2024], combining the local feature extraction capabilities of CNNs with the global context modeling of transformers. By integrating the strengths of both architectures, these hybrid models have shown enhanced performance in various medical imaging tasks, leading to better generalization, improved efficiency, and greater robustness in clinical applications.

Luo et al. [2021] incorporated a transformer-based module within the bottleneck of a traditional CNN-based encoder-decoder framework. This integration allowed the network to capture both local and global dependencies, enhancing feature representation. Similarly, Feng et al. [2022] introduced an innovative module consisted of two convolutional layers followed by a transformer module. This hybrid design enabled more effective feature learning by blending the hierarchical feature extraction power of CNNs with the self-attention mechanism of transformers. Another model proposed a dual-path architecture where a UNet module operated in parallel with a transformer block Soh et al. [2023]. This design aimed to maximize the advantages of both architectures, ensuring that the network could efficiently process high-resolution spatial details while maintaining a broad contextual understanding. Further advancements incorporating hybrid CNN-transformer architectures adopted advanced components to refine boundary estimations, effectively addressing challenges associated with fuzzy and non-smooth boundaries—an issue particularly relevant in stroke segmentation tasks Wu et al. [2023b,a].

One of the primary challenges in developing new architectures for stroke diagnosis is the limited availability of training data. Stroke features exhibit significant variations, such as infarct location, number, morphology, and size. To capture these diverse features, large and complex deep learning models have been developed. However, these models require substantial datasets to avoid overfitting, further exacerbating the data scarcity issue. This underscores the need to design lightweight models that incorporate various aspects of stroke into their pipelines to ensure robustness and efficiency Soliman et al. [2024]. Another critical issue in current research is the lack of external datasets for validation and testing. This limitation hinders the ability to assess a model's generalization performance on unseen data, which is essential for real-world clinical applications. Without robust external validation, the adaptation of these models in clinical settings remains challenging, limiting their practical utility and reliability.

3.1.3 Uni-Modality vs. Multi-Modality

While the majority of research in deep learning has traditionally focused on utilizing a single modality as input, multi-modal approaches have recently received increasing attention due to their ability to integrate diverse sources of information. These approaches leverage multiple data types, such as neuroimaging scans, biosignals, and clinical records, enabling models to capture a more comprehensive representation of complex medical conditions Jain and Srivastava [2024]. By incorporating heterogeneous data, multi-modal deep learning systems have the potential to enhance diagnostic accuracy, improve decision-making, and align more closely with real-world clinical workflows.

Furthermore, multi-modal data can either originate from entirely different sources or be derived from the same modality in ways that highlight distinct features. A notable example of the latter is the use of inductive biases, particularly the bilateral quasi-symmetry property of the brain Liang et al. [2021], Clèrigues et al. [2020]. This property suggests that brain structures exhibit near-symmetry across hemispheres, a characteristic that can be exploited to generate additional training data. In such studies, each brain slice is flipped and registered onto the original slice before being processed by a deep model. Significant deviations from the expected symmetry can indicate abnormalities, aiding in more accurate diagnosis. However, this approach has drawbacks, including the additional registration step, which increases computational complexity and may limit its practicality in real-world applications.

One of the primary limitations in deep multimodal learning for stroke diagnosis is computational complexity and scalability. High-dimensional neuroimaging data, such as 4D CT Perfusion and multimodal

MRI, require extensive computational resources, making real-time stroke prediction challenging. Studies using deep neural networks with CTP imaging have shown improved outcome predictions over conventional approaches, but the significant computational burden restricts deployment in emergency clinical settings Amador et al. [2025]. Moreover, training deep learning models on such large-scale datasets demands significant GPU resources, limiting accessibility in resource-constrained hospitals. To mitigate this, lightweight models such as MobileNet and federated learning approaches have been suggested to distribute computation across multiple institutions while preserving data privacy Borsos et al. [2024].

Beyond computational concerns, data heterogeneity and modality imbalance are major obstacles. Stroke datasets often vary in imaging protocols, patient demographics, and data quality, leading to model bias. A systematic review on multimodal machine learning for stroke diagnosis and prognosis highlighted the challenges of inconsistent datasets, particularly in integrating structured clinical data with unstructured imaging and biosignals Shurrab et al. [2024]. Another study examined DeepStroke, a multimodal adversarial deep learning framework for emergency stroke screening, which successfully fused facial video data, speech patterns, and imaging but struggled with domain adaptation when applied across different hospital settings Cai et al. [2022].

Another pressing issue is the fusion strategy used in multimodal learning. Combining different modalities—such as MRI, CT, EEG, and clinical metadata—requires an optimal fusion technique. Existing approaches include early fusion (concatenating raw data features), late fusion (combining individual model predictions), and hybrid fusion (integrating feature representations at intermediate layers). A study on Dynamic Affine Feature Map Transform (DAFT) demonstrated superior performance by adaptively scaling and shifting feature maps based on tabular patient data, leading to improved functional outcome prediction Borsos et al. [2024]. However, fusion methods remain highly dependent on dataset quality and pre-processing steps, limiting their generalizability across institutions.

The real-world deployment of deep multimodal stroke models is another significant barrier. While AI-driven approaches have shown state-of-the-art performance in retrospective studies, integrating them into clinical workflows requires seamless compatibility with hospital information systems. Many studies fail to address operational challenges such as data standardization, interoperability with electronic health records, and compliance with regulatory frameworks Cai et al. [2022]. Additionally, stroke diagnosis is time-sensitive, and current deep learning models require extensive inference times for accurate and robust diagnosis, which may not be practical in emergency settings. Efforts are underway to develop real-time, edge-based AI models that can be deployed in stroke triage systems, such as DeepStroke, a multimodal adversarial deep learning framework for emergency stroke screening. Despite these challenges, the future of deep multimodal learning in stroke care remains promising. Recent studies are exploring self-supervised learning techniques to reduce dependence on labelled datasets, adopting new segmentation approaches, enabling AI models to learn meaningful representations from unlabeled multimodal stroke data Samak et al. [2025], Rahman et al. [2025]. Additionally, federated learning frameworks Elhanashi et al. [2024], Bhatt et al. [2024] are being developed to facilitate collaborative AI model training across multiple institutions without compromising patient privacy. Moreover, the adoption of explainable AI techniques, such as models based on causal inference, may enhance clinician confidence in the predictions made by deep learning systems for stroke White et al. [2024a], Moulaei et al. [2024]. A comparative analysis of a selection of related approaches is presented in Table 2.

Table 2: Summary of key multimodal deep learning studies in stroke diagnosis and prognosis.

Source	Summary	Main Findings	Limitations	Outcome Measured
Shurrab et al. [2024]	Systematic review of multimodal machine learning methods for stroke prognosis and diagnosis, analyzing data modalities, methodologies, and challenges.	- Multimodal models outperform unimodal models in accuracy. - Strong evidence supports multimodal models in clinical practice. - Future research includes deep learning, federated learning, and multi-tasking models.	- Focused only on methodological components, omitting patient demographics. - Limited source selection may exclude relevant studies. - Manual data extraction may introduce errors.	Stroke prognosis, stroke diagnosis.
Raj et al. [2024]	Proposes a multilevel multimodal framework for automated collateral scoring in stroke.	- Achieved 91.17% accuracy for TAN score prediction and 94.12% for modified TAN scores. - Efficient and scalable for real-world use. - Attention maps enhance explainability.	- Needs validation with a larger, diverse dataset. - Data annotation could improve with more raters.	Accuracy and performance metrics, including F1 scores (0.86, 0.95) for multi-class and binary-class collateral scores.
Bhattacharya et al. [2020]	Enhances a multimodal stroke dataset and optimizes a deep neural network using the Antlion algorithm.	- Rigorous preprocessing addressed dataset issues. - Antlion Optimization (ALO) improved DNN hyperparameter selection. - DNN-ALO model outperformed others in accuracy, sensitivity, and specificity.	- Small dataset and data quality concerns. - Data imbalances from multi-institutional collection.	Training time efficiency, reducing training time to 38.13%.
White et al. [2024a]	Evaluates deep learning models for predicting post-stroke language deficits using MRI and tabular data.	- Top-performing 2D ResNet model achieved highest accuracy, AUC, and F1 score. - Model performed well on moderate/severe aphasia cases. - Combining imaging and tabular data improved classification.	- Limited to research-quality MRI scanners. - Crude measure of initial severity. - CNN complexity reduces interpretability, requiring explainable AI.	Spoken picture description score from Comprehensive Aphasia Test (CAT), classifying aphasia severity.
Jiang et al. [2023]	Proposes a deep learning-based multimodal MRI model for stroke patient nutritional risk assessment.	- 90% of Group A and 86.6% of Group B were at nutritional risk. - Enteral + parenteral nutrition (Group B) improved nutritional indicators, reduced infection/mortality rates. - Group B infection rate (36.6%) was half of Group A (76.6%).	- Limited test sites and materials. - Further research needed on endocrine metabolism and long-term monitoring.	Nutritional risk assessment, effectiveness of different nutritional support methods on clinical indicators and prognosis.
Amador et al. [2025]	Introduces a deep learning model combining 4D CT perfusion imaging and clinical metadata via cross-attention fusion.	- Intermediate fusion model outperformed late fusion for functional stroke outcome prediction. - 4D CTP + clinical metadata performed better than unimodal models. - A comprehensive view improves reliability.	- Small sample size and complexity of 4D CTP data. - Limited interpretability of interactions between modalities. - Data imbalance restricts fine-grained prediction.	Functional outcome at 90 days post-stroke (mRS score).
Cai et al. [2022]	Proposes DeepStroke, a deep learning framework using multimodal data (facial video and speech) for stroke screening.	- Outperformed traditional stroke triage and trained ER clinicians, achieving 10.94% higher sensitivity and 7.37% higher accuracy. - Screening result in under 6 minutes, much faster than MRI. - Multimodal data enhances stroke detection.	- Small sample size. - Errors in facial motion tracking due to 2D limitations. - Heavy computational workload affects mobile deployment.	Accuracy in detecting stroke presence in emergency room patients.
Borsos et al. [2024]	Presents a multimodal deep learning model (DAFT) for predicting acute ischemic stroke functional status using clinical data and CT perfusion imaging.	- DAFT model outperformed state-of-the-art methods (75% AUC, 80% F1 score). - Demonstrates feasibility of AI in predicting ischemic stroke outcomes. - TabNet model also performed well.	- Small sample size. - Limited CT perfusion spatio-temporal analysis. - Outcome prediction dichotomization limits result granularity.	Functional status of ischemic stroke patients, measured by modified Rankin Scale (mRS).

3.2 Prognosis

Deep learning plays an important role in stroke prognosis by providing data-driven insights into various aspects of the condition. These include predicting functional outcomes Jo et al. [2023], assessing the risk of hemorrhagic transformation or stroke recurrence, evaluating treatment response Hilbert et al. [2019], and enabling risk stratification. To achieve this, diverse data sources—such as imaging modalities, clinical records, and time-series data—can be leveraged Shurrab et al. [2024]. Different deep learning architectures are employed to process these data types effectively; for instance, RNNs are well-suited for sequential data, while CNNs excel at spatial feature extraction. By uncovering meaningful patterns, deep learning models can enhance prognostic accuracy and support more informed clinical decision-making. Stroke lesion segmentation plays a vital role not only in diagnosis but also in prognosis. Deep learning models enable precise quantification and localization of brain injuries from imaging modalities such as CT and MRI, allowing for accurate measurement of lesion volumes and assessment of tissue damage—both essential for predicting functional outcomes and guiding treatment decisions Wong et al. [2022]. For example, in Liu et al. [2023], a deep learning model was employed to fuse information from DWI images and clinical data from the acute phase to predict the ordinal 90-day modified Rankin Scale (mRS) score in acute ischemic stroke patients. The results demonstrated the high performance of this model in both internal and external test cohorts. Similarly, Fang et al. [2022] utilized deep learning and traditional machine learning models for six-month ischemic stroke outcome prediction on a public dataset. Ma et al. [2024] employed non-contrast computed tomography images and diagnostic reports, utilizing transformer-based architectures to predict functional outcomes, finding that multimodal data achieved superior results. Automated segmentation also facilitates monitoring lesion evolution over time, offering insights into treatment efficacy. The adoption of automated lesion segmentation not only enhances diagnostic precision but also supports large-scale research and clinical trials by providing standardized, high-quality data.

Other research has focused on predicting the final lesion outcome to guide treatment. In Nielsen et al. [2018b], deep learning models were used to predict the final imaging outcome by integrating acute imaging features, aiming to provide personalized treatment recommendations based on the volume of recoverable tissue. Similarly, Yu et al. [2020] pursued a comparable objective by predicting the size and location of infarct lesions 3 to 7 days after baseline, without prior knowledge of subsequent reperfusion status. The findings from these studies highlight the potential of deep learning models for individualized infarct lesion prediction in patients with acute ischemic stroke before any intervention is administered, paving the way for more targeted and effective therapies.

3.3 Rehabilitation

Deep learning is transforming stroke rehabilitation by automating the measurement of rehabilitation doses and functional recovery. Wearable sensors, such as Inertial Measurement Units (IMUs), combined with CNNs, enable real-time detection of movement primitives during therapy Kaku et al. [2020]. They developed a CNN model with instance normalization to classify functional primitives in stroke patients, achieving 70% accuracy in segmenting rehabilitation activities. This approach addresses the challenge of objectively quantifying rehabilitation doses, which is essential for optimizing neuroplasticity-driven recovery. Similarly, deep learning systems have been applied to detect compensatory movements in upper limb rehabilitation, providing therapists with actionable feedback to refine training protocols White et al. [2024b]. Clinical applications of deep learning extend to personalized rehabilitation strategies. A randomized trial compared conventional therapy with a deep learning-guided early intervention model for hemiplegic stroke patients. The intervention group showed statistically significant improvements in Fugl-Meyer Assessment (FMA) scores and Barthel Index compared to controls, underscoring the efficacy of data-driven rehabilitation protocols Khoramdel et al. [2021]. These systems often leverage temporal CNNs or recurrent architectures to analyze longitudinal motor performance data, enabling dynamic adjustments to therapy plans Senadheera et al. [2024], Xuefei et al. [2021]. Moreover, the integration of deep learning with emerging technologies such as virtual reality and robotic-assisted therapy is creating adaptive, feedback-driven rehabilitation systems. Such systems can continuously monitor patient performance in real time and adjust the difficulty of exercises to suit individual recovery levels, which fosters engagement and optimizes the intensity of therapy White et al. [2024b], Nielsen et al. [2018a].

Despite this progress, challenges in interpretability and real-world deployment remain. To improve interpretability, techniques such as feature importance analysis are increasingly used to validate that model decisions align with clinical expertise White et al. [2024b]. Future directions include federated learning to pool data across institutions and reinforcement learning for adaptive telerehabilitation systems Lin et al. [2022]. Furthermore, progress in deep learning for post-stroke rehabilitation is constrained by two main factors: the scarcity of large, standardized datasets and the necessity for interpretable models that clinicians can confidently integrate into their practice Sale et al. [2018], Mainali et al. [2021]. Overcoming these obstacles is key to unlocking the potential of deep learning to transform post-stroke rehabilitation through more precise, personalized, and effective therapeutic interventions.

4 Limitations and Challenges

There are several limitations and gaps in the adoption of full automated AI models into healthcare in general and in stroke in particular, including small data sizes, lack of external validation, and potential biases in models after training. These challenges underscore the need for future research to focus on larger and more diverse populations, along with robust and advanced validation methods, and new approaches to model training for stroke, to ensure that proposed AI models are generalizable and reliable for clinical practice. Some of the important challenges are detailed below.

4.1 Data Scarcity

Data scarcity is a major obstacle to employing deep learning-based models in healthcare applications. Collecting medical data is expensive, subject to strict privacy regulations, and often requires expert annotation, making large-scale dataset creation challenging. This limitation hinders the widespread development and evaluation of deep models, particularly in stroke care. In recent years, efforts have been made to create publicly available datasets to facilitate research and development in this field. However, the number of available samples in these datasets remains insufficient for large-scale training and validation. As a result, deep models trained on limited data often struggle with generalization, performing poorly on unseen cases. Such small datasets may fail to capture the full spectrum of disease presentation, including rare stroke subtypes or atypical lesion locations, leading to models that are not robust in real-world clinical scenarios. Additionally, such constraints can lead to biases, where models become overly tailored to specific conditions, demographics, or imaging protocols present in the training data. Addressing these challenges requires strategic approaches beyond simple data collection. Advanced techniques such as data augmentation, domain adaptation, and the integration of synthetic data generation or federated learning are crucial to improving model robustness and fairness. Furthermore, fostering multi-institutional collaborations to create larger, more diverse datasets is essential for training models that can generalize across different patient populations and healthcare systems.

4.2 Bias in Models

Beyond the limitations of data availability, which can introduce biases toward specific conditions, the training methodologies used for deep models also contribute to bias. In stroke lesion segmentation, a critical task for both diagnosis and prognosis, most models rely on loss functions such as Dice loss or its combinations with other functions like binary cross-entropy. However, these loss functions often fail to fully capture the complexity of stroke lesions, limiting the models' ability to learn optimally Kofler et al. [2023]. A significant issue arises in cases where infarcts vary in size. Standard loss functions tend to prioritize larger infarcts while overlooking smaller ones, introducing a bias toward detecting only more extensive stroke regions. Even standardized approaches that consider lesion size may struggle when multiple lesions of different sizes appear in the same slice or image, often overlooking smaller infarcts in such cases. This problem is compounded by biases originating from the data collection process itself, where certain patient demographics or stroke severities might be underrepresented, leading the model to perform poorly for minority groups. Additionally, detecting small infarcts is particularly challenging because their features can be subtle and easily confused with image noise or normal anatomical variations, especially if the model's feature extraction capabilities are not sufficiently sophisticated. To mitigate these biases, it is not enough to simply refine loss functions; research must also focus on developing more advanced network architectures and training strategies that are explicitly designed to handle class imbalance and feature ambiguity. Furthermore, integrating prior anatomical knowledge into the model can help constrain its predictions, reducing the likelihood of misclassifying healthy tissue. Employing multi-modal approaches and leveraging complementary information from different data sources can also enhance segmentation performance, leading to more reliable and equitable models.

4.3 Generalization

Variations in patient demographics, disease conditions, imaging protocols, and hardware systems can introduce significant discrepancies in input data for AI-based models. These variations can greatly impact the accuracy and reliability of AI predictions, particularly when tested on new, unseen data. One of the key challenges in medical AI is that real-world data often falls Out Of Distribution (OOD) of the training dataset.

This phenomenon, known as domain shift, can occur when a model trained on data from one hospital's MRI scanners is applied to images from another hospital using different hardware or acquisition parameters. This discrepancy reduces model robustness and limits its ability to generalize across diverse clinical scenarios. Moreover, many current studies validate models using a subset of the same dataset used for training, which may not provide an accurate assessment of real-world performance as the test data likely shares the same underlying distribution. To ensure AI models are truly generalizable, training processes must incorporate strategies to handle OOD data effectively. Employing domain adaptation techniques, such as adversarial training, self-supervised learning, or meta-learning, can improve model robustness. It is also imperative to move beyond internal validation and test models on large-scale, clinically diverse external datasets from multiple institutions. Ultimately, the gold standard for evaluation should be prospective clinical trials that assess model performance in real time, providing the most rigorous test of its clinical utility and generalizability.

4.4 Transparency of AI Models

One of the main challenges in deploying AI models in clinical settings is their "black box" nature, particularly in deep learning models, which often lack transparency in how they arrive at their final decisions. This opacity raises concerns among medical professionals, as the inability to interpret AI-driven diagnoses or predictions can hinder trust and widespread adoption. To ensure AI is effectively integrated into clinical workflows, it is essential to provide explanations for how models identify specific areas of the brain as stroke-related. This need for transparency extends to multiple stakeholders, including clinicians who must trust the output to make treatment decisions, patients who deserve to understand the basis of their diagnosis, and regulatory bodies that require evidence of safety and efficacy. Enhancing interpretability would not only help build confidence but also facilitate the identification and correction of errors, ensuring greater accountability. Various explainability techniques, such as saliency maps or attention-based visualizations, can highlight influential regions in an image. However, more advanced methods are emerging, such as generating counterfactual explanations ("what-if" scenarios) or extracting human-readable rules from the model. Integrating human-in-the-loop approaches, where AI systems and clinicians work collaboratively, can further refine and validate model interpretations. By prioritizing explainability and making it a core component of model design, AI can evolve from a black box to a trusted decision-support tool in medical practice.

5 Future Directions and Conclusion

The integration of deep learning into stroke care is poised to revolutionize clinical workflows, improving the speed and accuracy of diagnosis, prognosis, and rehabilitation. Future advancements should focus on refining deep learning models to enhance their interpretability, reliability, and adaptability across diverse patient populations and healthcare settings. Common challenges with AI such as the generalizability and explainability are major obstacles to wide-scale implementation in clinical use.Generalizability is now considered actively through foundation models that are still in the early stage of development and require several years to reach the stage of stable and trusted technology. Explainability remains a key concern, as clinicians must understand AI-driven recommendations to ensure trust and facilitate informed decision-making. The development of hybrid models that combine CNNs with transformer-based architectures can further enhance diagnostic performance by leveraging both local and global contextual features. Additionally, multimodal learning approaches, which integrate imaging data with EHRs, biosignals, and genetic markers, hold great promise for a more comprehensive assessment of stroke patients. However, challenges such as data scarcity, variability in imaging protocols, and privacy concerns must be addressed through standardized data-sharing frameworks and federated learning models that allow collaborative AI training without compromising patient confidentiality. Furthermore, real-time AI-assisted stroke detection systems, potentially embedded in mobile or point-of-care devices, could bridge the gap between early diagnosis and timely intervention, particularly in resource-limited settings.

As deep learning continues to evolve, its application in stroke rehabilitation is expected to become more personalized and adaptive, driven by real-time patient feedback and wearable sensor technologies. AI-powered rehabilitation tools, combined with virtual and augmented reality, could enable precise, engaging, and patient-specific therapy regimens that accelerate recovery and improve long-term outcomes. Reinforcement learning models, capable of dynamically adjusting treatment strategies based on patient progress, could lead to highly effective personalized rehabilitation programs. With better understanding of medical data associated with time-sensitive incidents such as stroke, there is a large potential that portable devices such as smart watches would enable alerting of potential stroke in an efficient way. Despite these promising advancements, ethical and regulatory challenges remain, necessitating clear guidelines for AI integration into clinical practice. Transparency, fairness, and bias mitigation strategies must be prioritized to ensure that AI-driven stroke care benefits all patient demographics equitably. Future research should focus on validating AI models through large-scale, multi-institutional trials to ensure generalizability and clinical robustness. Ultimately, the synergy between AI, medical expertise, and innovative rehabilitation strategies will pave the way for a more efficient, accurate, and accessible stroke care paradigm, significantly reducing the burden of this debilitating condition worldwide.

References

Valeriia Abramova, Albert Clerigues, Ana Quiles, Deysi Garcia Figueredo, Yolanda Silva, Salvador Pedraza, Arnau Oliver, and Xavier Lladó. Hemorrhagic stroke lesion segmentation using a 3d u-net with squeeze-and-excitation blocks. *Computerized Medical Imaging and Graphics*, 90:101908, 2021.

S Mazdak Abulnaga and Jonathan Rubin. Ischemic stroke lesion segmentation in CT perfusion scans using pyramid pooling and focal loss. In *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: 4th International Workshop, BrainLes 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 16, 2018, Revised Selected Papers, Part I 4*, pages 352–363. Springer, 2019.

Ramsha Ahmed, Aamna Al Shehhi, Naoufel Werghi, and Mohamed L Seghier. Segmentation of stroke lesions using transformers-augmented mri analysis. *Human Brain Mapping*, 45(11):e26803, 2024.

Fatima Alshehri and Ghulam Muhammad. Ischemic stroke segmentation by transformer and convolutional neural network using few-shot learning. *ACM Transactions on Multimedia Computing, Communications and Applications*, 20(12):1–21, 2024.

K. Amador et al. A cross-attention-based deep learning approach for predicting functional stroke outcomes using 4d ctp imaging and clinical metadata. *Medical Image Analysis*, 99:103381, 2025.

Abhishek Bal et al. A robust ischemic stroke lesion segmentation technique using two-pathway 3D deep neural network in MR images. *Multimedia Tools and Applications*, pages 1–40, 2023.

H. Bhatt et al. Artificial neural network-driven federated learning for heart stroke prediction in healthcare 4.0 underlying 5g. *Concurrency and Computation: Practice and Experience*, 36(3):e7911, 2024.

S. Bhattacharya et al. Antlion re-sampling based deep neural network model for classification of imbalanced multimodal stroke dataset. *Multimedia Tools and Applications*, pages 1–25, 2020.

B. Borsos et al. Predicting stroke outcome: a case for multimodal deep learning methods with tabular and ct perfusion data. *Artificial Intelligence in Medicine*, 147:102719, 2024.

T. Cai et al. Deepstroke: An efficient stroke screening framework for emergency rooms with multimodal adversarial deep learning. *Medical Image Analysis*, 80:102522, 2022.

Carlo W Cereda, Søren Christensen, Bruce CV Campbell, Nishant K Mishra, Michael Mlynash, Christopher Levi, Matus Straka, Max Wintermark, Roland Bammer, Gregory W Albers, et al. A benchmarking tool to evaluate computer tomography perfusion infarct core predictions against a dwi standard. *Journal of Cerebral Blood Flow & Metabolism*, 36(10):1780–1789, 2016.

Danton S Char, Nigam H Shah, and David Magnus. Implementing machine learning in health care—addressing ethical challenges. *New England Journal of Medicine*, 378(11):981–983, 2018.

Soren Christensen and Maarten G Lansberg. Ct perfusion in acute stroke: practical guidance for implementation in clinical practice. *Journal of Cerebral Blood Flow & Metabolism*, 39(9):1664–1668, 2019.

Albert Clèrigues, Sergi Valverde, Jose Bernal, Jordi Freixenet, Arnau Oliver, and Xavier Lladó. Acute and sub-acute stroke lesion segmentation from multimodal mri. *Computer methods and programs in biomedicine*, 194:105521, 2020.

Alexandra L Czap and Sunil A Sheth. Overview of imaging modalities in stroke. *Neurology*, 97 (20_Supplement_2):S42–S51, 2021.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

A. Elhanashi et al. Telestroke: real-time stroke detection with federated learning and yolov8 on edge devices. *Journal of Real-Time Image Processing*, 21(4):121, 2024.

Andre Esteva, Brett Kuprel, Roberto A Novoa, Justin Ko, Susan M Swetter, Helen M Blau, and Sebastian Thrun. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639): 115–118, 2017.

Gang Fang, Zhennan Huang, and Zhongrui Wang. Predicting ischemic stroke outcome using deep learning approaches. *Frontiers in genetics*, 12:827522, 2022.

Valery L Feigin, Emma Nichols, Tahiya Alam, Michele S Bannick, Ettore Beghi, Natacha Blake, William J Culpepper, E Ray Dorsey, Alexis Elbaz, Florian Fischer, et al. Global, regional, and national burden of neurological disorders during 1990–2015: a systematic analysis for the global burden of disease study 2015. *The Lancet Neurology*, 16(11):877–897, 2017.

Valery L Feigin, Michael Brainin, Bo Norrving, Sheila Martins, Ralph L Sacco, Werner Hacke, Marc Fisher, Jeyaraj Pandian, and Patrice Lindsay. World stroke organization (wso): global stroke fact sheet 2022. *International Journal of Stroke*, 17(1):18–29, 2022.

Pan Feng, Bo Ni, Xiantao Cai, and Yutao Xie. Utransnet: Transformer within u-net for stroke lesion segmentation. In *2022 IEEE 25th International Conference on Computer Supported Cooperative Work in Design (CSCWD)*, pages 359–364. IEEE, 2022.

Anjali Gautam and Balasubramanian Raman. Towards effective classification of brain hemorrhagic and ischemic stroke using cnn. *Biomedical Signal Processing and Control*, 63:102178, 2021.

MJ Graves. Magnetic resonance angiography. *The British Journal of Radiology*, 70(829):6–28, 1997.

Arsany Hakim, Søren Christensen, Stefan Winzeck, Maarten G Lansberg, Mark W Parsons, Christian Lucas, David Robben, Roland Wiest, Mauricio Reyes, and Greg Zaharchuk. Predicting infarct core from computed tomography perfusion in acute ischemia with machine learning: Lessons from the isles challenge. *Stroke*, 52(7):2328–2337, 2021.

Moritz R Hernandez Petzsche, Ezequiel de la Rosa, Uta Hanning, Roland Wiest, Waldo Valenzuela, Mauricio Reyes, Maria Meyer, Sook-Lei Liew, Florian Kofler, Ivan Ezhov, et al. Isles 2022: A multi-center magnetic resonance imaging stroke lesion segmentation dataset. *Scientific data*, 9(1):762, 2022.

Adam Hilbert, Lucas Alexandre Ramos, Hendrikus JA van Os, Sílvia D Olabarriaga, Manon L Tolhuisen, Marieke JH Wermer, Renan Sales Barros, Irene van der Schaaf, Diederik Dippel, YBWEM Roos, et al. Data-efficient deep learning of radiological image data for outcome prediction after endovascular treatment of patients with acute ischemic stroke. *Computers in biology and medicine*, 115:103516, 2019.

Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7132–7141, 2018.

Fabian Isensee, Paul F Jaeger, Simon AA Kohl, Jens Petersen, and Klaus H Maier-Hein. nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature methods*, 18(2): 203–211, 2021.

S. Jain and R. Srivastava. Multi-modality nde fusion using encoder–decoder networks for identifying multiple neurological disorders from eeg signals. *Technology and Health Care*, 2024.

Y. Jiang et al. Nutritional risk assessment and countermeasures for stroke patients based on deep learning and multimodal mri images. *International Journal of Computational Intelligence Systems*, 16(1):72, 2023.

Hongju Jo, Changi Kim, Dowan Gwon, Jaeho Lee, Joonwon Lee, Kang Min Park, and Seongho Park. Combining clinical and imaging data for predicting functional outcomes after acute ischemic stroke: an automated machine learning approach. *Scientific reports*, 13(1):16926, 2023.

Aakash Kaku, Avinash Parnandi, Anita Venkatesan, Natasha Pandit, Heidi Schambra, and Carlos Fernandez-Granda. Towards data-driven stroke rehabilitation via wearable sensors and deep learning. In *Machine Learning for Healthcare Conference*, pages 143–171. PMLR, 2020.

J Khoramdel, A Moori, MM Moghaddam, and E Najafi. Compensatory movement detection in upper limb rehabilitation with deep learning methods. In *2021 9th RSI International Conference on Robotics and Mechatronics (ICRoM)*, pages 465–471. IEEE, 2021.

Bum Joon Kim, Hyun Goo Kang, Hye-Jin Kim, Sung-Ho Ahn, Na Young Kim, Steven Warach, and Dong-Wha Kang. Magnetic resonance imaging in acute ischemic stroke treatment. *Journal of stroke*, 16(3):131, 2014.

Florian Kofler et al. Panoptica–instance-wise evaluation of 3D semantic and instance segmentation maps. *arXiv preprint arXiv:2312.02608*, 2023.

Amish Kumar et al. CSNet: A new deepnet framework for ischemic stroke lesion segmentation. *Computer Methods and Programs in Biomedicine*, 193:105524, 2020.

Maarten G Lansberg, Alex M Norbash, Michael P Marks, David C Tong, Michael E Moseley, and Gregory W Albers. Advantages of adding diffusion-weighted magnetic resonance imaging to conventional magnetic resonance imaging for evaluating acute stroke. *Archives of neurology*, 57(9):1311–1316, 2000.

Kongming Liang et al. Symmetry-enhanced attention network for acute ischemic infarct segmentation with non-contrast CT images. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part VII 24*, pages 432–441. Springer, 2021.

Michelle P Lin and David S Liebeskind. Imaging of ischemic stroke. *Continuum: Lifelong Learning in Neurology*, 22(5):1399–1423, 2016.

Ping-Ju Lin, Xiaoxue Zhai, Wei Li, Tianyi Li, Dandan Cheng, Chong Li, Yu Pan, and Linhong Ji. A transferable deep learning prognosis model for predicting stroke patients' recovery in different rehabilitation trainings. *IEEE Journal of Biomedical and Health Informatics*, 26(12):6003–6011, 2022.

Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen AW van der Laak, Bram van Ginneken, and Clara I Sánchez. A survey on deep learning in medical image analysis. *Medical image analysis*, 42:60–88, 2017.

Ruihua Liu et al. Pool-UNet: Ischemic stroke segmentation from CT perfusion scans using poolformer UNet. In *2022 6th Asian Conference on Artificial Intelligence Technology (ACAIT)*, pages 1–6. IEEE, 2022.

Yongkai Liu, Yannan Yu, Jiahong Ouyang, Bin Jiang, Guang Yang, Sophie Ostmeier, Max Wintermark, Patrik Michel, David S Liebeskind, Maarten G Lansberg, et al. Functional outcome prediction in acute ischemic stroke using a fused imaging and clinical deep learning model. *Stroke*, 54(9):2316–2327, 2023.

Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021.

Chun Luo, Jing Zhang, Xinglin Chen, Yinhao Tang, Xiechuan Weng, and Fan Xu. Ucatr: Based on cnn and transformer encoding and cross-attention decoding for lesion segmentation of acute ischemic stroke in non-contrast computed tomography images. In *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, pages 3565–3568. IEEE, 2021.

Danqing Ma, Meng Wang, Ao Xiang, Zongqing Qi, and Qin Yang. Transformer-based classification outcome prediction for multimodal stroke treatment. In *2024 IEEE 2nd International Conference on Sensors, Electronics and Computer Engineering (ICSECE)*, pages 383–386. IEEE, 2024.

Mohamed Mabrok, Yalda Zafari-Ghadim, Mostafa Mabrok, and Essam A Rashed. On the challenges and perspectives of stroke segmentation using ai. In *2024 IEEE International Conference on Future Machine Learning and Data Science (FMLDS)*, pages 129–134. IEEE, 2024.

S. Mainali, M. E. Darsie, and K. S. Smetana. Machine learning in action: stroke diagnosis and outcome prediction. *Frontiers in Neurology*, 12:734345, 2021. doi: 10.3389/fneur.2021.734345.

Adam Marcus, Paul Bentley, and Daniel Rueckert. Concurrent ischemic lesion age estimation and segmentation of ct brain using a transformer-based network. *IEEE Transactions on Medical Imaging*, 42(12): 3464–3473, 2023.

K. Moulaei et al. Explainable artificial intelligence for stroke prediction through comparison of deep learning and machine learning models. *Scientific Reports*, 14(1):31392, 2024.

A. Nielsen, M. B. Hansen, A. Tietze, and K. Mouridsen. Prediction of tissue outcome and assessment of treatment effect in acute ischemic stroke using deep learning. *Stroke*, 49(7):1394–1401, 2018a. doi: 10.1161/STROKEAHA.117.019740.

Anne Nielsen, Mikkel Bo Hansen, Anna Tietze, and Kim Mouridsen. Prediction of tissue outcome and assessment of treatment effect in acute ischemic stroke using deep learning. *Stroke*, 49(6):1394–1401, 2018b.

Ozan Oktay et al. Attention u-net: Learning where to look for the pancreas. *arXiv preprint arXiv:1804.03999*, 2018.

David B Olawade, Nicholas Aderinto, Aanuoluwapo C David-Olawade, Eghosasere Egbon, Temitope Adereni, Mayowa Racheal Popoola, and Ritika Tiwari. Integrating ai-driven wearable devices and biometric data into stroke risk assessment: A review of opportunities and challenges. *Clinical Neurology and Neurosurgery*, page 108689, 2024.

Nicholas G Perry, Benjamin B Warner, Chia-Ying Hsu, Yu-Wei Hsieh, Yung-Chieh Lin, Yu-Lun Wang, Yuan-Yuan Li, Jen-Kuang Lee, Po-Hsun Yang, Wen-Yih Huang, et al. Deep learning for stroke management: A review. *Frontiers in Neurology*, 11:666, 2020.

A. Rahman et al. Deep learning-driven segmentation of ischemic stroke lesions using multi-channel mri. *Biomedical Signal Processing and Control*, 105:107676, 2025.

R. Raj et al. Multilevel multimodal framework for automatic collateral scoring in brain stroke. *IEEE Access*, 2024.

P. Sale, G. Ferriero, L. Ciabattoni, A.M. Cortese, F. Ferracuti, and L. Romeo. Predicting motor and cognitive improvement through machine learning in post-stroke rehabilitation. *Journal of Stroke Cerebrovascular Diseases*, 27:2962–2972, 2018. doi: 10.1016/j.jstrokecerebrovasdis.2018.06.021.

Z. A. Samak et al. Automatic prediction of stroke treatment outcomes: latest advances and perspectives. *Biomedical Engineering Letters*, pages 1–22, 2025.

Isuru Senadheera, Prasad Hettiarachchi, Brendon Haslam, Rashmika Nawaratne, Jacinta Sheehan, Kylee J Lockwood, Damminda Alahakoon, and Leeanne M Carey. Ai applications in adult stroke recovery and rehabilitation: a scoping review using ai. *Sensors (Basel, Switzerland)*, 24(20):6585, 2024.

S. Shurrab et al. Multimodal machine learning for stroke prognosis and diagnosis: A systematic review. *IEEE Journal of Biomedical and Health Informatics*, 2024.

Claus Z Simonsen, Mette H Madsen, Marie L Schmitz, Irene K Mikkelsen, Marc Fisher, and Grethe Andersen. Sensitivity of diffusion-and perfusion-weighted imaging for diagnosing acute ischemic stroke is 97.5%. *Stroke*, 46(1):98–101, 2015.

Wei Kwek Soh, Hing Yee Yuen, and Jagath C Rajapakse. Hut: Hybrid unet transformer for brain lesion and tumour segmentation. *Heliyon*, 2023.

Ahmed Soliman, Yalda Zafari-Ghadim, Yousif Yousif, Ahmed Ibrahim, Amr Mohamed, Essam A Rashed, and Mohamed Mabrok. Deep models for stroke segmentation: do complex architectures always perform better? *IEEE Access*, 2024.

KJ Van Everdingen, J Van der Grond, LJ Kappelle, LMP Ramos, and WPTM Mali. Diffusion-weighted magnetic resonance imaging in acute stroke. *Stroke*, 29(9):1783–1790, 1998.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

A. White et al. Predicting recovery following stroke: Deep learning, multimodal data and feature selection using explainable ai. *NeuroImage: Clinical*, 43:103638, 2024a.

Adam White, Margarita Saranti, Artur d'Avila Garcez, Thomas MH Hope, Cathy J Price, and Howard Bowman. Predicting recovery following stroke: Deep learning, multimodal data and feature selection using explainable ai. *NeuroImage: Clinical*, 43:103638, 2024b.

Kelvin K Wong, Jonathon S Cummock, Guihua Li, Rahul Ghosh, Pingyi Xu, John J Volpi, and Stephen TC Wong. Automatic segmentation in acute ischemic stroke: Prognostic significance of topological stroke volumes on stroke outcome. *Stroke*, 53(9):2896–2905, 2022.

Zelin Wu, Xueying Zhang, Fenglian Li, Suzhe Wang, Lixia Huang, and Jiaying Li. W-net: A boundary-enhanced segmentation network for stroke lesions. *Expert Systems with Applications*, page 120637, 2023a.

Zelin Wu, Xueying Zhang, Fenglian Li, Suzhe Wang, and Jiaying Li. Transrender: a transformer-based boundary rendering segmentation network for stroke lesions. *Frontiers in Neuroscience*, 17, 2023b.

Zelin Wu, Xueying Zhang, Fenglian Li, Suzhe Wang, and Jiaying Li. A feature-enhanced network for stroke lesion segmentation from brain mri images. *Computers in Biology and Medicine*, 174:108326, 2024.

H Xuefei, Yongmei Yan, et al. Study on the effect of rehabilitation therapy based on deep learning on functional recovery of stroke patients with hemiplegia. *Frontiers in Medical Science Research*, 3(1):22–27, 2021.

Weihao Yu et al. Metaformer is actually what you need for vision. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10819–10829, 2022.

Yannan Yu, Yuan Xie, Thoralf Thamm, Enhao Gong, Jiahong Ouyang, Charles Huang, Soren Christensen, Michael P Marks, Maarten G Lansberg, Gregory W Albers, et al. Use of deep learning to predict final ischemic stroke lesions from initial magnetic resonance imaging. *JAMA network open*, 3(3):e200772–e200772, 2020.

Yalda Zafari-Ghadim, Essam A Rashed, Amr Mohamed, and Mohamed Mabrok. Transformers-based architectures for stroke segmentation: A review. *Artificial Intelligence Review*, 57(11):307, 2024.

Hengshuang Zhao et al. Pyramid scene parsing network. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2881–2890, 2017.